Case Study Report

Johan Nino Espino

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Introduction

In this case study, I looked at simulations of tiny particles, such as water droplets in clouds for example, moving through turbulent air. So understanding how these particles come together to form clusters is very important as it can help explain things like how rain forms. The data set that I worked on contains 89 simulation runs in which each run involves three input variables. These variables include the Reynolds number (Re) which measures how turbulent the flow is, the Froude number (Fr) which tells how strong gravity is, and the Stokes number (St) which measures how much inertia the particles have. And for each simulation, I analyzed the distribution of cluster volumes by looking at four statistical measures, which includes mean, variance, skewness, and kurtosis. From my understanding, the mean tell us the average cluster volume, the variance measures how much the cluster volumes differ from the average, skewness describes whether the distribution is more spread out toward smaller or larger clusters, and kurtosis helps us understand how likely the distribution is to contain unusually small or large clusters.

With this case study, I got two goals in mind. My first goal is that I want to predict these four statistical measures for new set of combinations of Re, Fr, and St. My second goal is that I want to understand how each input variable affects the cluster volume distribution and whether these effects differ across the four statistical measures. To do this, I use linear regression models on the log scale, which allows for a nonlinear relationship with St and an interaction between Re and Fr. I then select my models using cross-validation to evaluate how well these models predict new data. With this approach, it allows me to understand the influence of each variable while also keeping the models relatively simple and accurate.

Methodology

Preparing the data

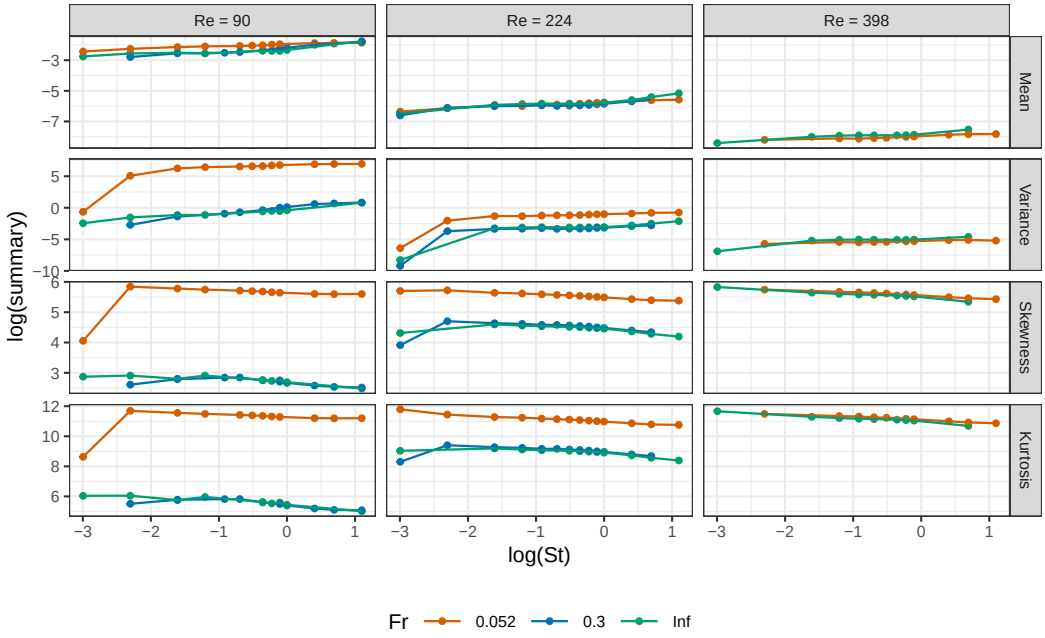
The simulations that were first provided to me has the first four raw moments of the cluster volumes, in which I use to calculate the mean, variance, skewness, and kurtosis using standard

statistical formulas such as $E[X^2] - (E[X]^2)$ to calculate the variance as an example.

I found that all four statistical measures do have a wide range of values. Like for the variance, it ranges from about 10^{-4} to 10^3, which means that some values are much larger than others. The variation also varies significantly between different simulation settings. So to account for these differences, I applied a log transformation to each measure before building my models. With this, it makes the data easier to analyze and it helps my models be better at understanding how each input variable influences the results.

For the input variables, I noticed that Re and Fr each have only three possible values. Thus I treated them as categorical variables. St has 14 different values in which most of them, bunched near 0, so I used log(St) to spread out these values and make it easier for me to understand how St affects the cluster volumes.

What I saw in the data



Here is **?@fig-eda** which gave me some clear insights.

The first thing that I noticed is that the relationship with St is not linear. From the graphs, even after I applied a log transformation to St, the relationship is not a straight line. Because of this, I tried to include polynomial terms in our models to better capture these nonlinear patterns.

The second thing that I see is that Re and Fr interact. When gravity is strong like Fr = 0.052, I see that the variance, skewness, and kurtosis have all increased significantly at Re =

90. But I do see that this increase becomes smaller at Re = 224 and is barely noticeable at Re = 398. What this could mean is that the effect of gravity does depends on the value of Re. So I included an interaction term between Re and Fr in our models to account for this relationship.

The third thing I noticed is that weak gravity has a similar effect to no gravity. I see that the lines for Fr = 0.3 and Fr = ∞ are almost identical. So I replaced Fr with a simpler gravity variable that has two categories. In this case, strong gravity (Fr = 0.052), and weak or no gravity (Fr = 0.3 or ∞). I find that this simpler approach predicts just as well as using all three Fr values while making the models easier to understand and interpret.